

1.3 Data recording, analysis and presentation	Learners should be able to demonstrate knowledge and understanding of the process and procedures involved in the collection, analysis and presentation of data and associated strengths and weaknesses from each of the four main research methods and techniques. This will necessitate the ability to perform some calculations (please see Appendix 5c for examples of mathematical requirements).
Raw data	• design of raw data recording tables • use of raw data recording tables • standard and decimal form • significant figures • make estimations from data collected.
Levels and types of data	• nominal level data • ordinal level data • interval level data • quantitative data • qualitative data • primary data • secondary data.
Descriptive statistics	• measures of central tendency ◦ mode, median, mean. • measures of dispersion ◦ variance, range, standard deviation. • percentages • fractions • frequency tables (tally chart) • line graph • pie charts • bar charts • histograms • scatter diagram.
Inferential statistics	• normal distribution curves • skewed distribution curves • probability • significance levels • using statistical tables of critical values • criteria for using a parametric test • criteria for using a specific non-parametric inferential test (Mann-Whitney U test, Wilcoxon T Signed Ranks test, Chi-square, Binomial Sign test and Spearman's Rho) • understand the use of specific non-parametric inferential tests (Mann-Whitney U test, Wilcoxon T Signed Ranks test, Chi-square, Binomial Sign test and Spearman's Rho)

	• type 1 errors • type 2 errors • symbols: =, <, >, <<, >>, ~, H_1, H_0, Σ, α, β.
Evaluative issues	• representativeness • generalisability • reliability: ◦ Internal, External, Inter-rater, Test-retest, Split-half • validity: ◦ Internal, Face, Construct, Concurrent, Criterion, External, Population, Ecological • demand characteristics • social desirability • researcher/observer bias • researcher/observer effect(s) • ethical considerations: ◦ informed consent, debrief, protection of participants, deception, confidentiality, right to withdraw.

Report writing
